

```
multiple_children.c
1#include <stdio.h>
2#include <unistd.h>
3#include <sys/wait.h>
4#include <stdlib.h>
5
6int main()
7{
8    int i;
9    pid_t pid;
10
11    for (i = 1; i <= 3; i++) {
12        pid = fork();
13        if (pid == 0) {
14            printf("Child %d: PID = %d\n", i, getpid());
15            sleep(i);
16            printf("Child %d finished\n", i);
17            exit(0);
18        }
19    }
20
21    printf("Parent: waiting for children...\n");
22
23    for (i = 0; i < 3; i++) {
24        pid = wait(NULL);
25        printf("Parent: Child with PID %d completed\n", pid);
26    }
27
28    printf("All children completed. Parent exiting.\n");
29
30    return 0;
31}
```

```
dell@DESKTOP-EFR7Q53: /mnt/c/Users/DELL/OneDrive/Desktop/OS/2520030283_Practical/Practical-04$ gcc multiple_children.c
dell@DESKTOP-EFR7Q53: /mnt/c/Users/DELL/OneDrive/Desktop/OS/2520030283_Practical/Practical-04$ ./a.out
Child 2: PID = 15713
Child 1: PID = 15712
Parent: waiting for children...
Child 3: PID = 15714
Child 1 finished
Parent: Child with PID 15712 completed
Child 2 finished
Parent: Child with PID 15713 completed
Child 3 finished
Parent: Child with PID 15714 completed
All children completed. Parent exiting.
dell@DESKTOP-EFR7Q53: /mnt/c/Users/DELL/OneDrive/Desktop/OS/2520030283_Practical/Practical-04$
```

```
zombie.c
1#include <stdio.h>
2#include <unistd.h>
3#include <stdlib.h>
4
5int main()
6{
7    pid_t pid = fork();
8
9    if (pid == 0) {
10        printf("Child: PID = %d\n", getpid());
11        printf("Child exiting...\n");
12        exit(0);
13    }
14    else {
15        printf("Parent: PID = %d\n", getpid());
16        printf("Parent sleeping...\n");
17
18        sleep(20);
19
20        printf("Parent exiting...\n");
21    }
22
23    return 0;
24}
```

```
dell@DESKTOP-EFR7Q53: /mnt/c/Users/DELL/OneDrive/Desktop/OS/2520030283_Practical/Practical-04$ gcc zombie.c
dell@DESKTOP-EFR7Q53: /mnt/c/Users/DELL/OneDrive/Desktop/OS/2520030283_Practical/Practical-04$ ./a.out
Parent: PID = 17833
Parent sleeping...
Child: PID = 17834
Child exiting...
Parent exiting...
```